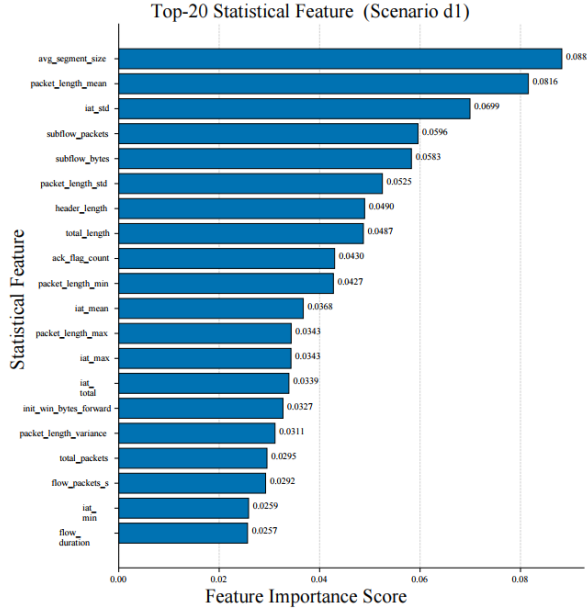
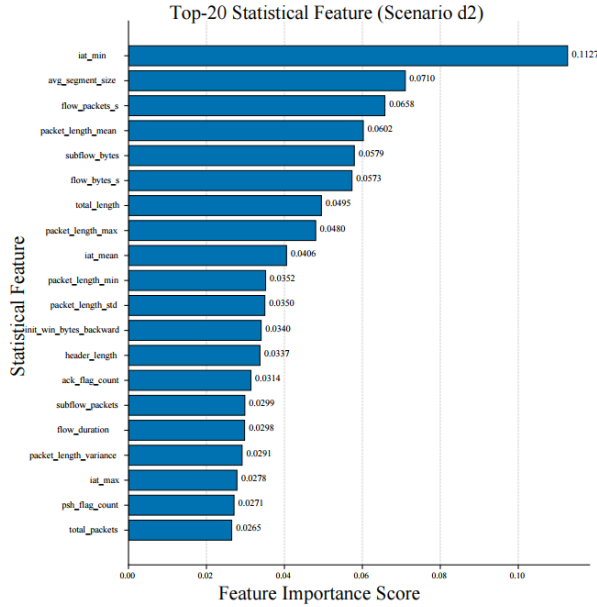


Fig. 12: RF feature importance on d_1 .Fig. 13: RF feature importance on d_2 .

VIII. APPENDIX

IX. RF FEATURE IMPORTANCE

The results of the static feature importance assessment are shown in the Fig. 12 and 13.

X. ADDITIONAL RESULTS

A. Additional Results on Different Strategies

The additional experiment results of different active learning strategies are shown in Tables VII-VIII.

TABLE V: Acquisition profile of the representative boundary case.

Item	Value
Dataset	USTC-TFC
Active-learning round	1
Sample ID	271200
Predicted label before query	Benign / label 0
Prediction probability before query	0.5236
Oracle label after query	Malicious / label 6
Uncertainty U	0.4764
Diversity S	1.0000
Density-aware support D	0.6543
DSM score	0.6869
Nearest benign distance	3.0814
Nearest malicious distance	0.5680
Boundary balance	0.3113
Boundary closeness	0.3540
Boundary score	0.1102

B. Additional Results on Trade-offs

The additional experiment results of performance and efficiency trade-offs are shown in Tables VI.

C. Additional Results on Different Layers

To analyze the trade-offs between TrafficBASS performance and computational cost, we evaluated ALBERT across different layer configurations under two scenarios. As shown in Table IX, deeper networks generally improve Acc but at the cost of higher computational demands. For all configurations, the 12-layer structure achieves a favorable balance between performance and efficiency. It achieves the lowest overall F1 (96.26%) on d_1 , while maintaining a reasonable prediction speed of 805 samples per second. This makes it well-suited for applications that require both high accuracy and moderate efficiency. Meanwhile, the 6-layer model performs slightly worse in terms of F1 (90.76%) but offers higher efficiency (1466 samples per second), making it a viable alternative for scenarios with stricter latency constraints or limited computational resources.

Although ALBERT employs parameter sharing across Transformer layers, computational complexity remains primarily influenced by forward propagation calculations. Raw traffic must traverse all Transformer layers sequentially. Computational demands increase linearly with layer count, extending processing time for deeper models. After Transformer processing, the resulting traffic representations feed into prediction components (typically fixed fully-connected layers or classifiers). These prediction components maintain consistent computational complexity regardless of model depth, resulting in similar traffic representation processing rates across models with varying layer counts.

TABLE VI: Performance and efficiency trade-offs on d_2 .

Base model	Fine-tuning	AL strategy	Acc(%)	F1(%)	Recall(%)	AUC	Speed	Param_To	Param-Ta	Time(s)
Pre-ALBERT	Lora	BASS	92.00	92.01	92.01	0.9935	790	11.8M	106K	10998
Pre-ALBERT	Adapter	BASS	86.21	86.21	85.79	0.9810	833	11.7M	45K	10500
Pre-ALBERT	Full	BASS	93.69	93.75	90.56	0.9967	914	11.7M	11.7M	10429
Random init	LoRA	BASS	68.53	66.77	68.21	0.9156	806	11.8M	105K	10913
Random init	Adaptor	BASS	65.39	63.58	65.39	0.9104	817	11.7M	45K	10691
Random init	Full	BASS	82.70	81.67	82.54	0.9771	935	11.7M	11.7M	10271

TABLE VII: Comparison of active learning strategies on scenario d_1 . "iters" represent iteration.

Strategy	F1(%)	iters to 80% F1	iters to F1
BASS	96.26	28	50
MAR	87.98	45	84
RAND	87.98	45	81
CONF	71.87	100	100
Diversity	87.21	49	95
Coreset	82.69	57	71
BALT	80.44	93	100
IMBL	85.64	75	98
ENT	77.65	100	100

TABLE VIII: Comparison of active learning strategies on scenario d_2 .

Strategy	F1(%)	iters to 80% F1	iters to F1
BASS	92.00	38	65
MAR	81.64	92	100
RAND	91.68	89	92
CONF	77.57	100	100
Diversity	90.35	68	92
Coreset	82.69	89	91
BALT	61.25	100	100
IMBL	77.65	80	95
ENT	81.64	100	100

In practical applications, the choice of model configuration depends on the specific requirements of accuracy and efficiency. 3-layer models prioritize speed and minimal resources at the cost of accuracy, suitable for real-time and resource-constrained environments. 6-layer models provide balanced performance for mid-range devices despite lower precision than deeper architectures. 12-layer models deliver optimal performance with reasonable resource usage, representing the best configuration for most production environments requiring high precision. 24-layer models prone to overfitting are unsuitable for lightweight tasks. However, with sufficient

TABLE IX: Compare with different layers of TraficBASS.

Layers	Size	d_1			d_2		
		Acc(%)	F1(%)	Speed	Acc(%)	F1(%)	Speed
3	41.2M	82.93	82.83	2481	68.19	65.7	2421
6	41.74M	90.98	90.76	1466	70.74	68.28	1432
12	42.8M	96.26	96.26	805	92.01	92.01	790
24	45.98M	84.92	84.76	468	52.11	44.23	414

data richness, this limitation can be mitigated, enabling their deployment in cloud-edge computing scenarios.

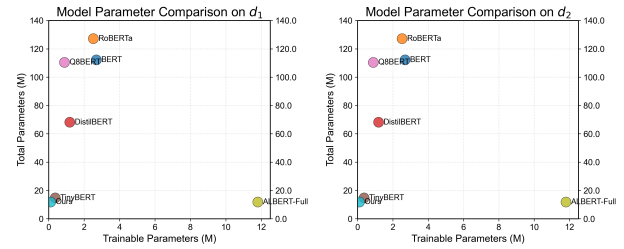
(a) Scenario d_1 (b) Scenario d_2

Fig. 14: Comparison of parameters on two scenarios.

D. Additional Results on Ablation study

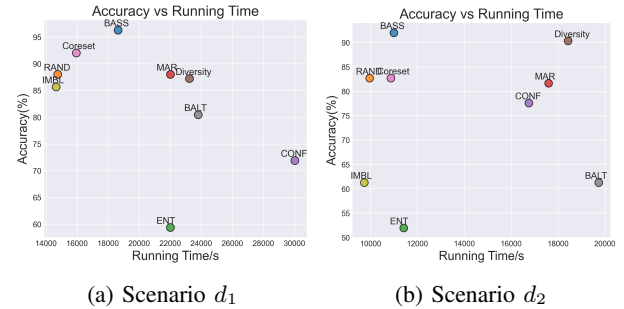
(a) Scenario d_1 (b) Scenario d_2

Fig. 15: Different AL strategies results on two scenarios.

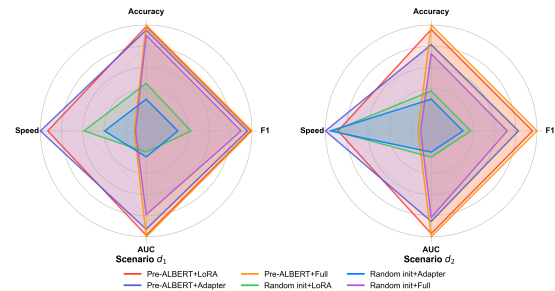


Fig. 16: Performance of different pre-training parameters and fine-tuning strategy on two scenarios.

E. Active Learning on Large Language Models

To further exploit the application of our framework on methods of LLMs, we compare the performance of TrafficBASS with the current LLMs models, including Phi3 [38] and Llama2 [39] from the general NLP domain, and CyBERTuned [40] from the cybersecurity domain with 7B-70B parameters. We adopted the same pre-training and fine-tuning strategy as TrafficBASS to observe the performance changes of LLMs over 100 iterations of boundary-aware active learning. The results are shown in Fig. 17, which depicts that CyBERTuned achieved only 29.89% and 27.84% F1 score despite extensive active learning (100 samples per round for 100 rounds) on two scenarios.

Notably, performance deteriorated as parameter count increased, indicating fundamental architectural incompatibility. This underperformance originates from architectural misalignment between NLP-optimized LLMs and structured network traffic patterns. The semantic gap creates a representational mismatch that fine-tuning cannot bridge with merely 10,000 samples, which is insufficient for models with billions of parameters. The temporal dependencies and evasive characteristics of APT traffic further challenge deep transformer architectures lacking network-specific inductive biases, indicating that domain-specific architectural adaptations, rather than parameter scaling, are essential for effective cybersecurity applications.

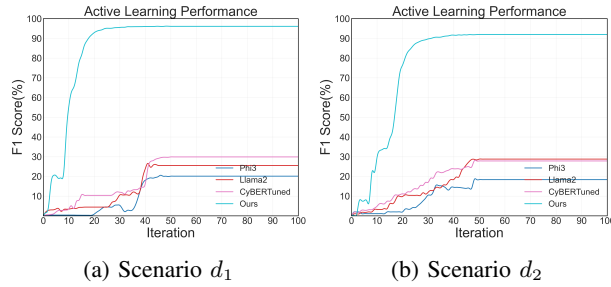


Fig. 17: Comparative F1 performance of LLMs during active learning on two scenarios.

F. Additional results on AEA robustness

1) *Perturbation Setting*: Given a normalized session representation $\mathbf{z}$ and its label y , a perturbed representation $\mathbf{z}'$ is constrained within an ℓ_∞ perturbation budget ϵ and clipped to the valid normalized feature range. The semantic label y is assumed unchanged for evaluation because the perturbation is used only as a feature-space stress test. No raw packets, payloads, protocol fields, or executable traces are modified.

2) *Perturbation-Budget Sensitivity*: To examine whether the robustness benefit of AEA remains stable under different perturbation strengths, we conduct a perturbation-budget sensitivity analysis using PGD-10. This experiment complements the main inference-time robustness evaluation by testing whether the observed feature-space robustness is tied to a single perturbation budget or persists across multiple budgets.

Setting. We compare TrafficBASS with its variant without AEA under PGD-10 perturbations with $\epsilon \in \{0.01, 0.02, 0.03, 0.05, 0.08\}$. All perturbations are applied to normalized session representations and clipped to the valid normalized feature range. The held-out test set is fixed across all perturbation budgets, and no perturbed test sample is used for active-learning acquisition or model updating. The clean F1 scores are 0.9684 and 0.9375 for TrafficBASS and TrafficBASS w/o AEA on d_1 , and 0.9201 and 0.8769 on d_2 , respectively.

Metrics. For each perturbation budget, we report PGD-10 F1, the absolute F1 drop from the corresponding clean F1, and the relative PGD drop (RPD):

$$\text{RPD} = \frac{F1_{\text{clean}} - F1_{\text{PGD}}}{F1_{\text{clean}}} \times 100\%. \quad (31)$$

TABLE X: Perturbation-budget sensitivity under PGD-10. F1 Drop is computed with respect to the corresponding clean F1 of each model.

Dataset	ϵ	TrafficBASS			w/o AEA		
		F1	Drop	RPD	F1	Drop	RPD
d_1	0.01	0.9256	0.0428	4.42	0.8535	0.0840	8.96
	0.02	0.9022	0.0662	6.84	0.8168	0.1207	12.87
	0.03	0.8675	0.1009	10.42	0.7536	0.1839	19.62
	0.05	0.8432	0.1252	12.93	0.7527	0.1848	19.71
	0.08	0.8142	0.1542	15.92	0.7359	0.2016	21.50
d_2	0.01	0.8986	0.0215	2.34	0.7928	0.0841	9.59
	0.02	0.8597	0.0604	6.56	0.7668	0.1101	12.56
	0.03	0.8187	0.1014	11.02	0.7029	0.1740	19.84
	0.05	0.8093	0.1108	12.04	0.6768	0.2001	22.82
	0.08	0.7475	0.1726	18.76	0.6557	0.2212	25.23

Results and analysis. As shown in Table X and Fig. 18, the F1 scores of both variants decrease as ϵ increases, which is expected under stronger adversarial stress tests. However, TrafficBASS consistently achieves higher PGD-10 F1 and lower RPD than TrafficBASS w/o AEA across all perturbation budgets on both datasets. On d_1 , AEA improves PGD-10 F1 by 0.0880 on average and reduces the average RPD by 6.43 percentage points. On d_2 , AEA improves PGD-10 F1 by 0.1078 on average and reduces the average RPD by 7.86 percentage points. Even under the largest perturbation budget $\epsilon = 0.08$, TrafficBASS maintains F1 scores of 0.8142 and 0.7475 on d_1 and d_2 , respectively, outperforming the w/o AEA variant by 0.0783 and 0.0918.

These results indicate that AEA improves local stability around the learned feature-space decision boundary

and that this benefit persists across different PGD-10 perturbation budgets. Nevertheless, this experiment is a normalized feature-space stress test. The perturbations do not modify raw packets, payloads, protocol fields, or executable attack traces. Therefore, the results should be interpreted as empirical feature-space robustness evidence rather than certified robustness against arbitrary packet-level adversarial traffic.

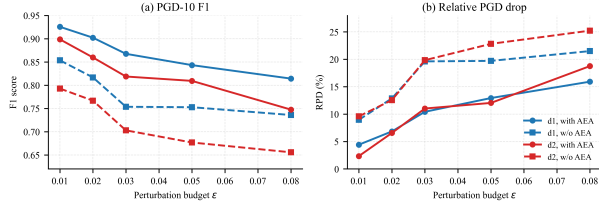


Fig. 18: Perturbation-budget sensitivity under PGD-10.

G. ALBERT with ALF-Guide Fine-tuning

ALBERT offers significant advantages for APT detection in traffic analysis. Its cross-layer parameter sharing reduces computational redundancy and enhances training stability. This efficiency is crucial when analyzing high-volume encrypted network flows with limited resources. Unlike BERT, ALBERT uses Sentence Order Prediction (SOP) instead of Next Sentence Prediction. SOP trains the model to distinguish between truly sequential samples and identical content in reversed order. This approach better captures temporal relationships in network traffic patterns. When combined with our ALF-driven LoRA fine-tuning methodology, ALBERT creates a powerful framework for APT detection. This integration preserves temporal reasoning capabilities while enabling rapid adaptation to emerging threats through targeted parameter updates.

H. Additional Results

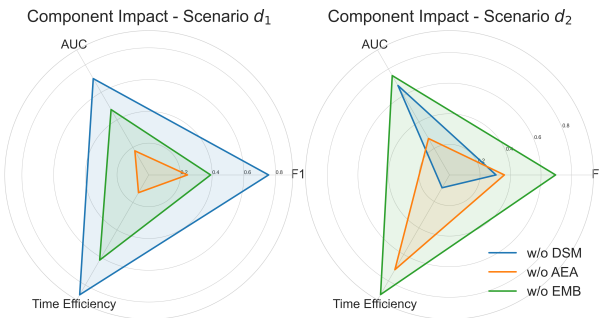


Fig. 19: Compare with different components of BASS.

TABLE XI: Comparison results with AL strategies.

Method	Iteration	d_1		d_2	
		F1(%)	AUC	F1(%)	AUC
RAND	20	42.3	0.9002	5.27	0.5581
	50	84.15	0.986	30.36	0.6903
	100	86.77	0.9887	80.50	0.9659
ENT	20	11.34	0.5335	11.62	0.6130
	50	50.42	0.6818	35.10	0.6903
	100	57.53	0.6947	44.23	0.8657
MAR	20	42.30	0.9002	18.78	0.7381
	50	84.15	0.9860	66.99	0.9327
	100	86.77	0.9887	81.29	0.9677
CONF	20	13.88	0.7978	3.41	0.5324
	50	50.48	0.9346	54.48	0.8834
	100	68.39	0.9593	76.77	0.9627
Diversity	20	36.63	0.8815	1.76	0.5988
	50	79.75	0.9755	48.15	0.9182
	100	87.01	0.9745	89.78	0.9835
Coreset	20	15.56	0.8218	5.27	0.6213
	50	39.48	0.9346	30.36	0.8834
	100	82.69	0.9753	80.50	0.9590
BALT	20	13.40	0.4938	2.00	0.5863
	50	36.60	0.6491	26.73	0.7166
	100	78.58	0.9871	55.80	0.8997
IMBL	20	19.66	0.7946	25.54	0.6564
	50	74.62	0.9607	42.52	0.8272
	100	84.94	0.9810	77.65	0.9612
BASS	20	55.44	0.9221	29.47	0.7377
	50	95.16	0.9941	87.32	0.9814
	100	96.26	0.9967	92.01	0.9935

TABLE XII: Inference-time robustness under adversarial perturbations. Both epsilon set to 0.03.

Scenario	Model	Clean F1	FGSM F1	PGD-10 F1	RPD (%)	RPG
d_1	ET-BERT ✓	0.9745	0.8453	0.7931	18.61	13.41%
	ET-BERT ×	0.9064	0.7384	0.6993	22.84	—
	Q8BERT ✓	0.9595	0.8032	0.7416	22.71	13.92%
	Q8BERT ×	0.8527	0.6173	0.6510	23.65	—
	TrafficBASS ✓	0.9684	0.8645	0.8675	10.42	15.11%
	TrafficBASS ×	0.9375	0.7440	0.7536	19.61	—
d_2	ET-BERT ✓	0.9118	0.7928	0.7602	16.62	22.61%
	ET-BERT ×	0.8429	0.6949	0.6200	26.44	—
	Q8BERT ✓	0.8914	0.7833	0.7820	12.27	11.99%
	Q8BERT ×	0.8607	0.6819	0.6983	18.86	—
	TrafficBASS ✓	0.9201	0.8112	0.8187	11.02	14.22%
	TrafficBASS ×	0.8769	0.7004	0.7029	19.84	—